

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed transfer pairs among Chapman–Shaoxing, CPSC2018, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (BCa, $B = 10,000$) on the full 60-experiment grid. The study followed a pre-registered protocol with dated, publicly archived amendments. The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60 (75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met (23/60 = 38.3%), triggering the registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (+0.0159 vs.

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+0.0083). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, although a source-fitted, label-free temperature predicts the *sign* of the benefit in 60/60 cells); and a discrimination-aware gate (C5) shows 7/12 pair $\times$ architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

For ECG this question is not academic. PTB-XL [3], Chapman-Shaoxing [4], and CPSC use different acquisition protocols, lead sets, and label vocabularies, so a model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [25]. Clinicians and regulators then face a concrete decision: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted under a convex loss, it minimizes NLL on the calibration split, so its calibration benefit there is empirically small; on a shifted target

the source-fit temperature is mismatched to the target logits and a non-trivial benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (nine negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing, protocol v2.1-A1 [26] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from decay to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [27, 28]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported.

Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, triggering the “ID nonzero boundary” branch (protocol A1.4.2): the

premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast restored as exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific. 47/60 = 78.3% of ID point estimates are positive and 2/60 = 3.3% have $\Delta\text{ECE}_{\text{ID}} < -0.01$ (counter-example 6, -0.0152 ; one marginal case, -0.0128).

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence, $n = 20,000$) recovers ground truth in the identifiable regime (27/27 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a threshold, and reports bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 methods $\times$ 6 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior adaptation is fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included (InceptionTime 15/30); after exclusion the EM OOD finding attenuates. The mechanism is prior-likelihood mismatch: EM fits a target prior that amplifies shift-induced miscalibration when the source confusion matrix is ill-conditioned.

C5 (Clinical, discrimination-aware calibration gate): A two-layer gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viability (accuracy $\geq$ source-domain baseline). Across 12 pair $\times$ architecture strata, 7/12 pass both layers; 5/12 fail the discrimination layer, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$, $\text{acc} = 0.657 < 0.713$). The

gate reframes the 9/60 counter-examples: 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures (C5 rejects despite C1 passing).

Prior ECG calibration studies report the phenomenon [12, 13, 20, 21] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty is *not* multi-seed validation on ECG (a methodological choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

2. Related Work

2.1. Calibration Estimation and Recalibration

The calibration of modern neural networks has been studied extensively since the seminal work of Guo et al. [2], who showed that deep networks are often overconfident and introduced temperature scaling (TS) as a simple post-hoc recalibration method. Nixon et al. [8] proposed the static calibration error, and Kumar et al. [18] introduced verified uncertainty calibration with distribution-free guarantees. Kull et al. [9] generalized TS to Dirichlet calibration for multi-class settings. In the clinical risk-model context, Van Calster et al. [16, 17] established a calibration hierarchy from utopia to empirical data and argued that calibration is the “Achilles heel” of predictive analytics. Strodtthoff et al. [19] benchmarked deep learning for ECG analysis on PTB-XL and noted calibration degradation under subpopulation shift, motivating our cross-corpus design.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

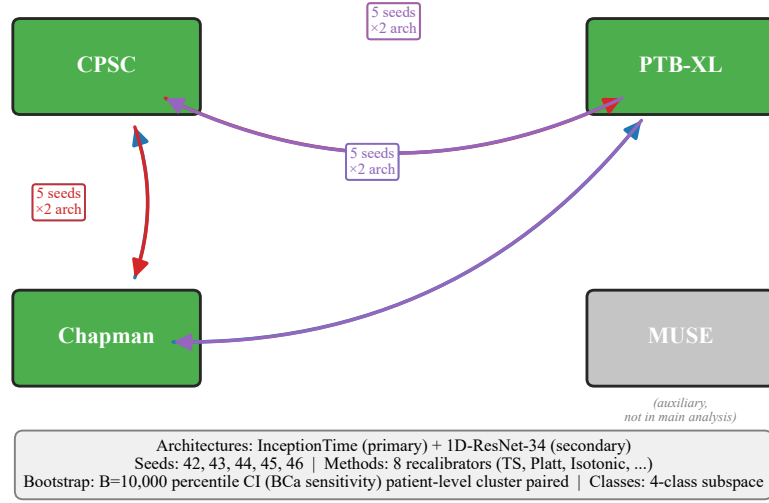


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, ResNet1D) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

120 2.2. *Prior-Shift Estimation and Transfer Calibration*

121 Lipton et al. [11] introduced black-box shift estimation (BBSE) for detect-
122 ing and correcting label shift using confusion-matrix inverses. Alexandari et al.
123 [10] showed that maximum-likelihood calibration with bias correction is hard-to-
124 beat at label shift adaptation, providing the theoretical foundation for our EM
125 prior-shift method. For transfer calibration upper bounds, TransCal [22] estimates
126 transferability for auto-selecting target-domain unsupervised optimal T ; Pseudo-
127 Cal [23] uses pseudo-labels for calibrated transfer; and LaSCal [24] provides label-
128 aware shift calibration with distribution-free guarantees. Our S1 paradigm differs
129 fundamentally: we fit on the *source* calibration split only (no target labels), with
130 S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly
131 weaker assumption than TransCal/PseudoCal, which access target logits or labels.

132 2.3. *Cross-Domain ECG Calibration*

133 Barandas et al. [12] evaluated uncertainty quantification methods in multi-
134 label ECG classification, finding that ensemble-based methods outperform single-
135 model approaches under subpopulation shift. Vranken et al. [13] studied uncer-
136 tainty estimation for 12-lead ECG deep learning and reported significant calibra-
137 tion drift across acquisition devices. Haekal et al. [20] performed cross-dataset val-
138 idation for atrial fibrillation detection with calibration-aware probability analysis,
139 and Patel & Beedala [21] documented calibration drift under cross-institutional
140 ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general
141 vision [1]; we contribute the multi-seed ECG quantification and the boundary-
142 condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

143 2.4. ECG Foundation Models and Transfer Learning (2024–2026)

144 Recent work has introduced ECG foundation models trained on millions of
145 recordings. ECGFounder [29, 37] (10 million ECGs, Harvard–Emory) reaches
146 expert-level performance on 80 diagnoses and transfers cross-domain via fine-
147 tuning. ECG-FM [30] is an open-weight transformer pretrained on 1.4 million
148 ECG segments with hybrid contrastive and generative self-supervision, showing
149 strong cross-dataset generalizability and label efficiency. HeartLang [31] treats
150 heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked
151 sentence pre-training. Tan et al. [32] found that fine-tuning benefits diminish
152 as the downstream dataset grows and that CNNs transfer more effectively than
153 RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD
154 [33] quantifies distribution-shift severity via uncertainty estimation to guide se-
155 lective LoRA layer unfreezing, and BeatRhythm-TTA [34] performs test-time
156 adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al.
157 [35] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-
158 preserving beat-wise augmentation for cross-source generalization, and Chen et al.
159 [36] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned
160 contrastive alignment. These works advance ECG transfer but not the *calibra-*
161 *tion* of transferred models, the gap our study fills via pre-registered cross-corpus
162 evaluation of recalibration methods.

163 3. Data and Pre-registered Protocol

164 Figure 1 summarizes the experimental design.

165 3.1. Datasets

- 166 • **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the down-
167 loaded `ptbx1_database.csv` copy contains 21,799 (a 38-record deficit in
168 the source CSV, disclosed as a provenance limitation), official 10-fold (folds
169 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records
170 are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- 171 • **Chapman-Shaoxing** [4, 7]: 20,265 raw records; 1 signal-read error
172 removed and 21 non-finite records quarantined (`qc_report.json`), for
173 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- 174 • **CPSC2018** [5, 6]: we use the union of the *CPSC Database* (6,877 records;
175 the public CPSC2018 training set) and the *CPSC-Extra Database* (3,453
176 records; the CPSC2018 recordings not used by the Challenge), both re-
177 leased through PhysioNet/CinC 2020. This is *not* a CPSC2018–2019 union:
178 CPSC2019 was a QRS/heart-rate *detection* task and defines no diagnos-
179 tic labels. Local extraction retains 6,844 of the 6,877 CPSC Database
180 records (33 records, A6845–A6877, were lost during archive extraction;
181 disclosed as a provenance limitation) plus all 3,453 CPSC-Extra records.
182 Patient-wise split. Two labeling facts must be stated explicitly. First,
183 CPSC2018’s official taxonomy is nine disease classes (Normal, AF, I-AVB,
184 LBBB, RBBB, PAC, PVC, STD, STE) with no superclass grouping; the
185 NORM/MI/STTC/CD superclass system we adopt is PTB-XL’s [3], ap-
186 plied here as a cross-corpus harmonization choice, not a CPSC-defined
187 one. Second, CPSC2018’s nine classes contain *no* MI; the MI class in our
188 4-class subspace is populated entirely by CPSC-Extra’s SNOMED anno-

tations (164867002, 164865005, 54329005). A strict official-class-name caliber would therefore reduce the subspace to {NORM, CD, STTC}; we retain MI because it is the only way to keep a myocardial-infarction category on the CPSC side of every transfer pair. HYP is extremely rare ($n = 11$) → main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides of the pair recomputed symmetrically.

3.2. *Label Mapping and Leakage Guard*

Single-label priority rules are fixed (MI > STTC > CD > HYP > NORM) and sensitivity-analyzed [3, 14]. Sinus rate/form statements (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard assertion in the reproducibility repo). Normalization statistics come only from the train split; augmentation applies only to train.

4. **Methods**

4.1. *Transfer pairs and shift matrix*

We study six directed cross-corpus transfer pairs: Chapman→CPSC, Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman, and PTB-XL→CPSC. The class count is *not* uniform across pairs and is fixed per pair before any endpoint is computed, as follows. A pair whose source or target is CPSC2018 is restricted to the 4-class subspace {NORM, CD, STTC, MI} (CPSC’s HYP class has $n = 11$, §3), with both sides of the pair recomputed symmetrically: this covers Chapman→CPSC, CPSC→Chapman, CPSC→PTB-XL,

212 and PTB-XL→CPSC ($K = 4$). The two pairs connecting Chapman–Shaoxing
213 and PTB-XL, which contain no CPSC side, retain the full 5-class vocabulary
214 {NORM, MI, STTC, CD, HYP}: Chapman→PTB-XL and PTB-XL→Chapman
215 ($K = 5$). Each experiment records its own K and label mapping (num_classes,
216 label_map in transfer_result.json), so no endpoint mixes class counts
217 within a cell. Consequently the primary endpoint is a top-label ECE on a 4- or
218 5-class task depending on the pair, and the pooled $\Delta\text{ECE}_{\text{OOD}}$ mixes both K values
219 across cells; the per-pair tables (Tables 1–4) are therefore the appropriate level
220 at which to read K -specific results. The L2 eval-time shift matrix (no retraining)
221 covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise
222 injection (BWL/MA/EM) at SNR {24, 12, 6, 0, −6} dB; gain $\times 0.5 / \times 2$.

223 4.2. Architecture and seeds

224 We use two architectures: InceptionTime (primary, 46,000–184,000 param-
225 eters) and ResNet1D (a 1D-ResNet-34 backbone) (secondary, with full main-
226 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The Bi-
227 Mamba architecture is included only as a toy experiment ($n = 60$ records, down-
228 sampled resolution) and deferred to the supplement; the full-resolution BiMamba
229 run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the
230 main-endpoint analysis or the architecture-robustness claim. Training: AdamW +
231 cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training;
232 temperature applied only as post-hoc temperature scaling.

233 4.3. Calibration methods and scenarios

234 We compare the no-recalibration baseline (None) against eight recalibration
235 methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scal-

ing, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and
BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline
plus these eight recalibration methods constitute the delivered set; CORAL, TTA,
MC-Dropout, and Oracle are not delivered on real data (a registered deviation,
see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply),
the main analysis scenario (Sec. 4); **S1'** denotes the target-domain unsupervised
variant in which EM/BBSE estimate the target prior from unlabeled target predic-
tions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [26]):

$$\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}} \quad (1)$$

where $ECE_{\text{raw}}^{\text{OOD}}$ is the 4-class-task top-label confidence SmoothECE on the tar-
get corpus with raw softmax probabilities, and $ECE_{\text{TS}}^{\text{OOD}}$ is the same after source-
cal-fit temperature scaling. Following the standard convention of Guo et al.
[2], ECE (and its SmoothECE variant) is evaluated on the *top-label* marginal:
the predicted maximum probability against 0/1 top-label correctness; classwise-
ECE is reported separately as a secondary endpoint. SmoothECE bandwidth:
 $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta ECE_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta ECE_{\text{OOD}} > 0 \quad (2)$$

The one-sided alternative $H_1 : \Delta ECE_{\text{OOD}} > 0$ follows the empirical asymmetry
registered at revision A1: on ID-like data the source-fit temperature yields, empir-
ically, little calibration change, whereas on a shifted target it is mismatched and a

257 non-trivial benefit is often retained [2, 1]. This direction prior is empirical, *not* a
 258 theorem: TS applied to an under-confident target can increase ECE, and our own
 259 data contain nine negative cells. The choice was made before the pilot results,
 260 which only triggered the endpoint re-location (revision A1). The one-sided test
 261 is the pre-registered choice, archived before multi-seed unblinding. Patient-level
 262 cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [15]; stratified pooling:
 263 strata = transfer pair $\times$ architecture.

264 The pre-registered CI method is BCa [15], at $B = 10,000$, re-run on the en-
 265 tire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The
 266 percentile bootstrap is the large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$;
 267 for the paired-cluster design with $B = 10,000$ and per-patient clusters, the dis-
 268 crepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were
 269 run ($B = 10,000$ percentile re-runs), BCa and percentile CIs agree in sign and
 270 magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain
 271 as historical provenance (5 lack log provenance, see Limitations); percentile CIs
 272 serve only as a sensitivity cross-check.

273 Boundary-condition endpoint:

$$\Delta \text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

274 The pre-registered expectation is that $H_0 : \Delta \text{ECE}_{\text{ID}} = 0$ is *not rejected*; the criterion
 275 is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid*
 276 *re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered*
 277 *(percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

278 Secondary endpoints: G (recoverable loss vs. oracle; *not delivered* — the Or-
 279 acle recalibrator was implemented (`fit_oracle/apply_oracle`) but never ex-
 280 ecuted on real data, so G is undefined here and is excluded from all reported

281 results; see Limitations), $\text{decay} = \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descrip-
282 tive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept,
283 predicted/true prior L_1 .

284 Family structure: BH-FDR $q = 0.05$ over $6 \text{ pairs} \times 2 \text{ arch} \times 13 \text{ shift levels} =$
285 156 primary comparisons; the multi-seed validation is robustness evidence (60
286 seed experiments), not a confirmatory family.

287 4.5. Pre-registration revision A1 (transparency statement)

288 The primary endpoint was re-located from decay to $\Delta\text{ECE}_{\text{OOD}}$ via pre-
289 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent
290 exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the
291 original ID-vs-OOD contrast. The revision direction is consistent with the empir-
292 ically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived
293 as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes.
294 The pilot results do not enter the main test report. All hypotheses in this paper are
295 tagged *exploratory + robustness validation*, not confirmatory.

296 The original BH-FDR family of 156 comparisons ($6 \text{ pairs} \times 2 \text{ architectures} \times$
297 13 shift levels) is reduced to a confirmatory family of 12 hypotheses ($6 \text{ pairs} \times 2$
298 $\text{architectures} \times 1 \text{ primary shift level}$) under amendment A2, with the remaining
299 13 shift levels designated as exploratory dose-response. Under the reduced family,
300 BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12
301 yields 2/12. The A2 amendment is archived in `docs/PROTOCOL_AMENDMENT_A2_`
302 `FAMILY_REDUCTION.md`.

303 4.6. *Pre-registration amendment A3 (corpus identity and MI provenance)*

304 A post-hoc audit of the CPSC corpus against the official challenge documen-
305 tation produced two wording corrections and one new disclosure, archived as
306 amendment A3 (docs/PROTOCOL_AMENDMENT_A3_CORPUS_IDENTITY.md). The
307 registered protocol described the third corpus as “CPSC2018+2019”; it is in fact
308 **CPSC2018 only**, comprising the CPSC Database (6,877 records) and the CPSC-
309 Extra Database (3,453 records). CPSC2019 was a QRS/heart-rate *detection* task
310 with no diagnostic labels and never entered the corpus. The new disclosure is that
311 CPSC2018’s official nine-class taxonomy contains no MI, so the MI class of our 4-
312 class subspace is populated entirely by CPSC-Extra’s SNOMED annotations; un-
313 der a strict official-class-name caliber the subspace would reduce to three classes.
314 A3 is a naming and attribution correction plus a disclosure: it changes *no* recorded
315 value, no split, and no analysis decision. The registered protocol snapshots are left
316 byte-identical and A3 is archived separately, so the before/after pair required by
317 pre-registration practice is preserved. The audit also found that 33 CPSC Database
318 records (A6845–A6877) were silently lost when the source archive was extracted
319 (6,877 in the archive vs. 6,844 on disk), disclosed as a provenance limitation.

320 4.7. *Decomposition of the benefit decay*

321 We decompose the retained benefit into slope / intercept / prevalence contribu-
322 tions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

323 where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the
324 chain residual (zero under nested counterfactuals). Identifiability is checked via

the Fisher information determinant on a pre-registered threshold τ ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

4.8. Two-layer joint bootstrap

Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$ joint Δ ECE CI (two-layer; honest non-nested approximation), per protocol §7.

4.9. Net Clinical Value (NCV)

Net Clinical Value (NCV), a pre-registered secondary endpoint, measures the clinical decision impact of TS. For each (sample, class) pair we compare the binary decision (threshold $\tau = 0.5$) before and after calibration, counting TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves clinical decisions. It is computed on ID and OOD splits (ncv_id, ncv_ood; E3 CSV).

4.10. Tooling

Parts of the analysis and plotting code were drafted and refactored with the assistance of a large language model assistant (GLM), used as a supervised coding assistant. All experiments and statistical analyses were executed with the author’s own scripts, and every reported number is reproducible from the public repository

(Code availability). The author reviewed the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

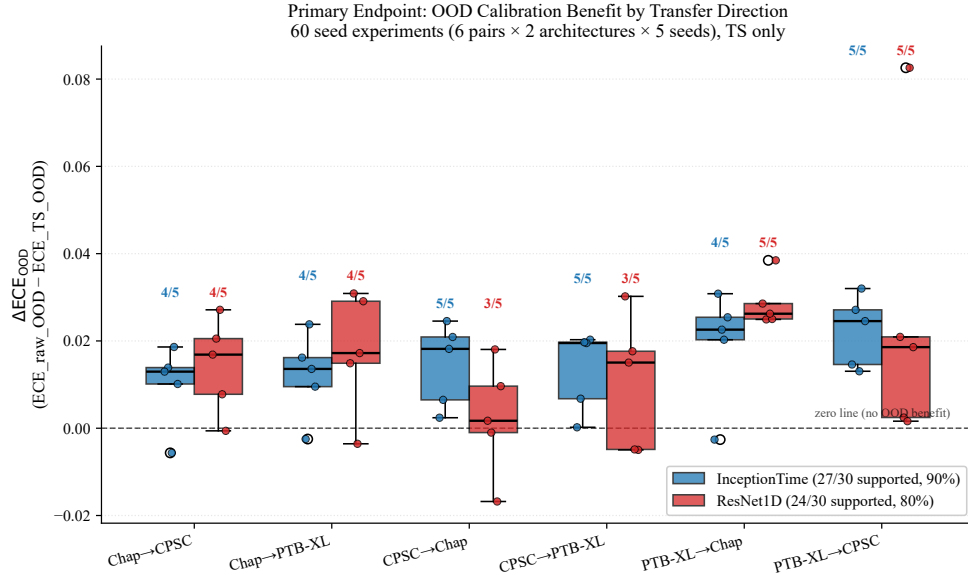


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. Class count $K = 4$ for the four pairs involving CPSC (Chap $\rightarrow$ CPSC, CPSC $\rightarrow$ Chap, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC) and $K = 5$ for the two CPSC-free pairs (Chap $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chap); see §4.1. “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap $\rightarrow$ CPSC ($K=4$)	IT	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	RN	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap $\rightarrow$ PTB-XL ($K=5$)	IT	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
	RN	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap ($K=4$)	IT	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	RN	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
CPSC→PTB-XL ($K=4$)	IT	42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓
		44	+0.0158	+0.0195	+0.0189	✓
		45	+0.0052	+0.0197	+0.0193	✓
		46	+0.0002	+0.0002	+0.0002	✓
	RN	42	+0.0114	+0.0150	+0.0148	✓
		43	+0.0048	−0.0050	−0.0051	✗
		44	−0.0032	−0.0049	−0.0049	✗
		45	+0.0130	+0.0176	+0.0173	✓
		46	+0.0261	+0.0302	+0.0296	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
PTB-XL→Chap ($K=5$)	IT	42	+0.0239	+0.0308	+0.0275	✓
		43	+0.0245	+0.0254	+0.0225	✓
		44	+0.0173	+0.0203	+0.0195	✓
		45	+0.0162	+0.0226	+0.0200	✓
		46	−0.0011	−0.0026	−0.0029	✗
	RN	42	+0.0145	+0.0249	+0.0191	✓
		43	+0.0338	+0.0385	+0.0336	✓
		44	+0.0213	+0.0250	+0.0228	✓
		45	+0.0228	+0.0262	+0.0231	✓
		46	+0.0239	+0.0286	+0.0221	✓
PTB-XL→CPSC ($K=4$)	IT	42	+0.0196	+0.0271	+0.0244	✓
		43	+0.0166	+0.0245	+0.0222	✓
		44	+0.0183	+0.0320	+0.0303	✓
		45	+0.0006	+0.0131	+0.0105	✓
		46	+0.0097	+0.0146	+0.0123	✓
	RN	42	+0.0010	+0.0025	+0.0023	✓
		43	+0.0197	+0.0186	+0.0145	✓
		44	+0.0635	+0.0826	+0.0772	✓
		45	+0.0001	+0.0016	+0.0015	✓
		46	+0.0126	+0.0209	+0.0190	✓

356 Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%,
357 ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence (10/10,
358 zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed 45). The
359 nine counter-examples are explained in Sec. 6.1.

360 The 60 experiments are balanced (6 directions $\times$ 2 architectures $\times$ 5 seeds, no
361 partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture interaction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5 on both architectures (10/10).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID} has mean $+0.0083$ (vs. OOD mean $+0.0159$, a $1.9\times$ ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast is restored as an exploratory endpoint; and C1 is

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

385 qualified, the OOD benefit being about $1.9\times$ the ID benefit rather than an exclu-
386 sively OOD effect. The positive ID estimates reflect finite-sample calibration-split
387 variation, SmoothECE estimator bias, and a val-ECE early-stopping effect (model
388 selection on the endpoint family), all disclosed.

389 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

390 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on Inception-
391 Time. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE achieves 22/30

392 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most
 393 fragile on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood
 394 mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60
 395 (85.0%) (InceptionTime 27/30, 90.0%; ResNet1D 24/30, 80.0%), with full 5-seed
 396 coverage on all 6 directions (Figure 4).

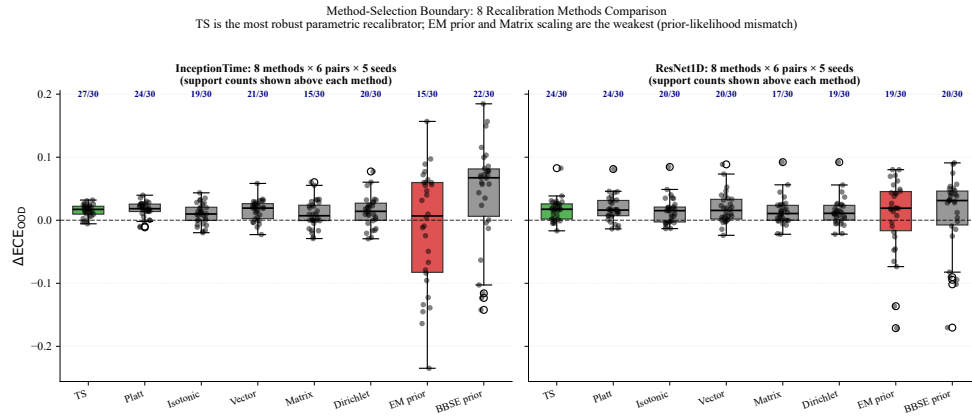


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

397 Across the 30 TS experiments per architecture, the standard deviation of
 398 ΔECE_{OOD} is 0.0098 (InceptionTime) and 0.0180 (ResNet1D), a $1.84\times$ ratio, with
 399 ResNet1D’s highest per-direction variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, in-
 400 cluding an unexcluded favorable outlier at seed 44, $\Delta ECE_{OOD} = +0.0826$). Sup-
 401 port deficits concentrate in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a
 402 direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields

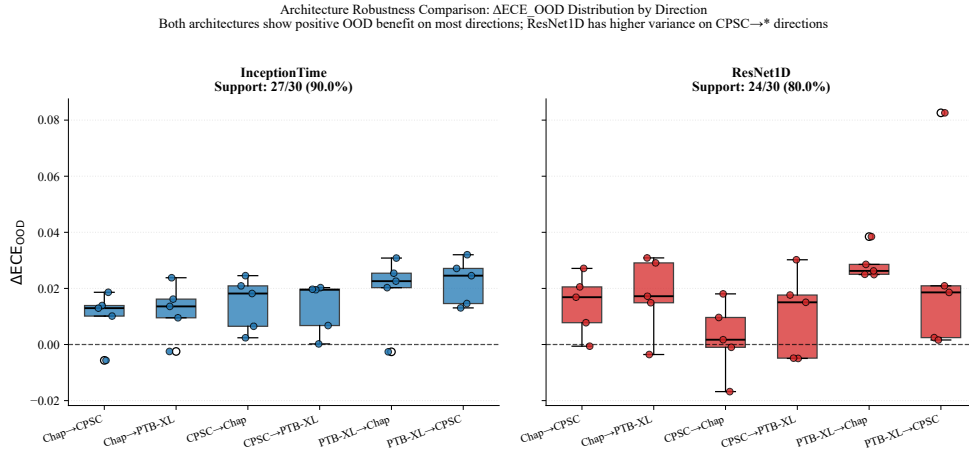


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (InceptionTime). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat*: 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

403 $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%)
 404 is not significant at $\alpha = 0.05$.

405 EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal,
 406 then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated
 407 confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies
 408 rather than corrects the shift-induced miscalibration. EM reaches only 15/30 sup-
 409 port on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-
 410 signature-specific), not method-intrinsic.

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

411 *Why TS and not Platt..* Platt achieves a marginally higher mean OOD benefit
412 than TS (+0.0176 vs. +0.0159; InceptionTime +0.0169 vs. +0.0152, Table 3),
413 which invites the question of why TS is singled out as the recommended default.
414 Three measured criteria separate them, and all three favour TS. First, *robustness*
415 *of the sign*: TS supports 51/60 cells versus Platt’s 48/60, i.e. TS fails to certify
416 a positive benefit in three fewer experiments, which is the quantity a deployment
417 decision actually depends on. Second, *dispersion*: TS has a smaller across-cell
418 standard deviation (0.0146 vs. 0.0167, a 13% reduction), and the methods with
419 the highest mean benefit are markedly less stable (BBSE +0.0266 at sd 0.0736,
420 EM −0.0026 at sd 0.0793) — a high mean bought with a heavy tail is the worst

421 profile for a per-deployment recalibrator. Third, *the difference is not statistically*
 422 *distinguishable*: the patient-level paired contrast over the 60 matched cells gives
 423 $\overline{\Delta ECE}_{\text{TS}} - \overline{\Delta ECE}_{\text{Platt}} = -0.0017$ (sd 0.0117, paired $t = -1.14$, $p \approx 0.26$), with
 424 TS the larger benefit in exactly 30/60 cells — a coin flip. The mean advantage of
 425 Platt is therefore inside the noise, whereas the support advantage of TS is not a
 426 noise-level quantity: it is a count of cells in which the sign is certified. Given that
 427 the two methods are statistically indistinguishable on magnitude, the tie is broken
 428 on certification rate and dispersion, both of which favour TS. A further structural
 429 point reinforces the choice: TS is a single-parameter monotone transform and
 430 therefore exactly argmax-preserving, whereas per-class Platt and isotonic are fit-
 431 ted independently per class, which generates each class’s score from a different
 432 monotone map and so does *not* preserve the ranking within a sample (the imple-
 433 mentation declares this explicitly in its docstring). TS is thus the only method
 434 in the delivered set whose calibration guarantee is obtained without perturbing
 435 the discrimination it leaves untouched (§5.9). None of this makes Platt wrong; it
 436 makes Platt a reasonable alternative whose mean advantage is not resolvable at
 437 $n = 60$, and whose certification rate is lower.

438 5.4. Shapley decomposition validation (27 cells, $n = 20,000$)

439 The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in$
 440 $\{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposi-
 441 tion recovers ground truth within the pre-registered identifiable regime (Fisher
 442 information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the
 443 recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of
 444 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in ex-
 445 treme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most

446 clinically significant (see `decomposition_validation.csv`; the $n = 20,000$
 447 CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three
 448 entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\det \in \{2.4 \times$
 449 $10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with
 450 the recovery-failed flag set. Bootstrap CIs on absolute Shapley values and share-
 451 reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

452 The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE;
 453 its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of
 454 decomposition explanatory power) remains to be run on real data; per the protocol
 455 self-binding clause, “mechanism” wording is limited to “mechanism decomposi-
 456 tion (synthetic validation part)” until cross-fitting completes.

457 5.5. Predictability experiment (*failure branch triggered*)

458 The predictability experiment (protocol §8.5) tests whether an unsupervised
 459 target-domain statistic (predicted-prior L_1 distance, logit moments) predicts ben-
 460 efit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The imple-
 461 mented fold unit is (architecture, transfer pair) (12 folds), versus the registered
 462 pair-only folds (6); under that registered deviation the pair-only recomputation re-
 463 tains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime +
 464 ResNet1D); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

465 Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$.
 466 The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5 , trig-
 467 gering the **failure branch**: the three-step chain (decomposition + predictability +
 468 deployment) is downgraded to a **two-step chain** (decomposition + deployment),
 469 and the deployment criterion is renamed “empirical lookup table” rather than “pre-
 470 dictive criterion”. No post-hoc predictability claim is made.

471 Re-running the LOO R^2 with three label-free features (fitted temperature T ,
 472 Wasserstein-1 distance between source and target logit distributions, class-wise
 473 entropy difference) gives a sharper and less favourable picture for the bound-
 474 ary. Only T carries signal: under the implemented (architecture, pair) folds it
 475 reaches LOO $R^2 = +0.340$ (bootstrap CI $[+0.044, +0.363]$), and under per-cell
 476 folds $+0.361$ (CI $[+0.201, +0.584]$), while the Wasserstein distance (-0.064) and
 477 the entropy difference (-0.033) carry none. T is also a perfect *directional* predic-
 478 tor: $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells, against 51/60 (85%) for the
 479 constant-positive baseline. We therefore do *not* claim that no label-free quantity
 480 carries information about the endpoint — T does. The claim we retain is the
 481 pre-registered one: no label-free feature reaches the LOO R^2 threshold of 0.5
 482 under the implemented fold unit ($0.363 < 0.5$), so the failure branch stands and
 483 the three-step chain remains downgraded to two steps. Two qualifications are
 484 required. First, the verdict is fold-unit-sensitive: per-cell folds give a CI upper
 485 bound of 0.584, which would *not* trigger the branch, so the downgrade should be
 486 read as a property of the coarser (architecture, pair) unit as much as of the data.
 487 Second, T 's predictive power is not independent evidence of a substantive bene-
 488 fit: $T = 1$ is a fixed point of the transform, so part of the relation is mechanical,
 489 and the sign rule is the direction temperature scaling must have by construction
 490 (Sec. 5.8). The gate in §5.9, based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination
 491 criterion, remains the appropriate actionable criterion.

492 5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

493 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
 494 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
 495 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	−0.096	−0.720	+0.104	Failure branch
Platt	−0.059	−0.584	+0.002	Failure branch
Isotonic	+0.011	−0.569	+0.101	Failure branch
Vector	−0.061	−0.522	+0.073	Failure branch
Matrix	−0.137	−0.516	−0.013	Failure branch
Dirichlet	−0.084	−0.394	+0.032	Failure branch
EM prior	+0.143	−0.250	+0.274	Failure branch
BBSE prior	+0.136	−0.354	+0.201	Failure branch

496 SmoothECE ≤ 0.05 and Δ ECE CI lower bound > 0 , primary-endpoint conven-
497 tion). Comparators: always-none, always-TS, always-buy- n -labels.

498 Sensitivity / specificity (Table 6): Youden’s J is maximized at raw ECE
499 $= 0.127$, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sensitivity is
500 high (92% of beneficial cases caught) but specificity low (20% of non-beneficial
501 cases excluded; 80% falsely flagged), reflecting the asymmetric cost structure:
502 missing a beneficial recalibration costs more than an unnecessary one. *Caveat*: the
503 threshold is searched on the same 1230-cell pool used for reporting (in-sample).
504 A leave-shift-type-out recomputation (threshold selected excluding the held-out
505 dose level, evaluated on it) yields $J = +0.17$ (downsampling), $J = -0.03$ (lead
506 drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127
507 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE > threshold → predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

508 four held-out dose levels (`results/deployment_holdout_recompute.csv`).
 509 The in-sample numbers are therefore optimistic, and the criterion requires inde-
 510 pendent validation before operational use.

511 Every lookup-table input is *audit-time* (raw ECE, ID–OOD accuracy gap,
 512 per-cell labels), requiring target ground-truth labels unavailable in the unlabeled
 513 stream. The only label-free proxies examined are the predictability experiment’s
 514 target statistics (predicted-prior L_1 distance, logit moments) plus the source-fitted
 515 temperature T ; that experiment triggered its pre-registered failure branch (LOO
 516 R^2 CI upper bound $0.104 < 0.5$, Table 5). T does carry signal (LOO $R^2 = +0.34$;
 517 sign of the endpoint correct in 60/60 cells), but it is a source-side quantity describ-
 518 ing the transform being evaluated rather than an independent target-side statistic
 519 (Sec. 5.8), so no label-free deployable criterion can be claimed. The lookup table
 520 is an *analysis- and audit-time decision aid* for a model owner with labeled eval-
 521 uation data; the actionable deployment path is small-sample recalibration: buy n
 522 labeled target records and recalibrate locally (protocol §5 S2). A deployable gate
 523 remains future work (Limitations xiii).

524 Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfy-
525 ing the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 ,
526 point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion
527 was not evaluated), a safety rate of 0.0064. Sign convention: $\Delta\text{ECE} = \text{post-}$
528 $\text{recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the
529 primary-endpoint convention. *Coverage note:* the L2 shift matrix has been com-
530 pleted on all 390 planned (pair, seed, shift) combinations ($6 \text{ pairs} \times 5 \text{ seeds} \times 13$
531 levels ; results/l2_shift_full_390cells.csv); the safety figures quoted here are from
532 the initial two-seed InceptionTime subset (seeds 42–43, ResNet1D L2 excluded),
533 so these numbers rest on 2-seed, single-architecture coverage. The criterion is
534 rarely satisfied, so TS is a useful default but not a universally safe recalibrator,
535 and should be combined with local validation.

536 At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 ver-
537 sus 0.0999 for treating all cells; by the 90th percentile it turns negative
538 (-0.0024). Net benefit over treat-all is modest, confined to low thresholds
539 (results/deployment_metrics.csv).

540 Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE
541 -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145)
542 on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the
543 target-label upper bound. The lookup table needs no target labels and captures
544 most of the TS benefit.

545 5.7. Noise floor control: SmoothECE differential under perfect calibration

546 A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact
547 of SmoothECE’s sensitivity to temperature scaling T under perfect calibration
548 ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated pre-

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155

549 ditions ($T_{\text{true}} = 1$) we apply T and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) -$
550 $\text{SmoothECE}(\text{raw})$.

551 The floor is strongly *distribution-dependent*: for flat or wide confidence
552 distributions it is essentially zero, or slightly negative at moderate T . We
553 therefore do not evaluate it on synthetic distributions. Instead we use the
554 empirical top-label confidence distribution of each of the 60 main-grid cells
555 (checkpoints/e2_probs_cache/*.npz), relabel each cell as perfectly cali-
556 brated by resampling $y \sim \text{Bernoulli}(c)$ at its own confidence c (three draws per
557 cell), and report the band across cells. This removes any freedom to choose a
558 distribution that suits the argument, and it evaluates the floor where the study’s
559 own predictions actually live (pooled median confidence 0.79, with 30% of sam-
560 ples above 0.9). The synthetic three-distribution sweep (uniform, normal-skewed,
561 bimodal at $n = 2,000$) is retained as a sensitivity analysis.

562 The noise floor grows monotonically in $|T - 1|$ and is distribution-dependent.
 563 Evaluated on the 60 main-grid cells — median across cells, with the cross-cell
 564 range in parentheses — $T = 1.0$ gives exactly 0 (exact cancellation); $T = 1.25$
 565 gives $+0.0138$ (-0.0115 to $+0.0294$); $T = 1.5$ gives $+0.0389$ (-0.0047 to
 566 $+0.0597$); $T = 3.0$ gives $+0.1260$ ($+0.0404$ to $+0.1605$); and $T = 4.5$ gives
 567 $+0.1582$ ($+0.0605$ to $+0.2129$). The synthetic sweep agrees in magnitude for
 568 high-confidence distributions ($T = 1.5$: $+0.039$ to $+0.055$) but not in sign for flat
 569 ones ($T = 1.25$: -0.009 for uniform), which is why the empirical distributions
 570 are used. The floor is positive in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and
 571 in 90% ($T = 1.25$) to 97% ($T = 1.5$) of cells in between. Observed raw ECE
 572 across the 62 raw-stage cells of the TS-component ablation under the corrected
 573 label encoding (`results/ablation_ts_components.OLD_ENCODING.csv`,
 574 `stage=raw`) ranges 0.036 to 0.357 (mean 0.211), i.e. $2\text{--}21\times$ the raw ECE under
 575 perfect calibration (≈ 0.017); the nine counter-examples also show raw ECE well
 576 above the floor (0.065–0.321 vs. ≈ 0.017), so they are not perfect-calibration
 577 cases. At their own fitted temperatures ($T \in [0.93, 1.00]$) the floor is ≤ 0.013 ,
 578 comparable to their $|\Delta\text{ECE}|$ (≤ 0.017), but its direction is opposite to theirs, so
 579 they are not floor artifacts; their common mechanism is given in Sec. 5.8. The
 580 direction of the floor is what matters for the endpoint, and it is fixed. In the
 581 primary convention ($\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}} - \text{ECE}_{\text{post}}$) the floor inflates ECE_{post}
 582 and therefore contributes *negatively*. The identity is the minimiser of ECE
 583 under perfect calibration, so the floor is positive in the noise convention on
 584 both sides of $T = 1$: in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and in 90–97%
 585 of cells for $1.1 \leq T \leq 1.5$. The idealised endpoint is therefore $-\text{floor}(T) \leq 0$
 586 almost everywhere, and a predominantly attenuating artefact cannot generate the

587 positive endpoint observed in 51/60 experiments. We test this and three further
588 implications of the floor hypothesis in Sec. 5.8.

589 5.8. Noise-floor mechanism test

590 The control above quantifies the floor; it does not by itself decide whether the
591 floor *drives* the endpoint. We therefore test four implications of the “floor artefact”
592 hypothesis, using the source-fitted temperature T recovered per cell from the TS-
593 component ablation under the corrected label encoding.

594 (i) *Direction..* The floor contributes negatively to $\Delta\text{ECE}_{\text{OOD}}$ in 90–100% of cells
595 at every $T \neq 1$ (Sec. 5.7), so the hypothesis predicts a systematically *negative*
596 endpoint. The observed mean is +0.0159, positive in 51/60 cells.

597 (ii) *Sign flip with T ..* A fixed-direction artefact cannot produce a sign that flips
598 with T . In the corrected data $T \in [0.932, 1.498]$ (median 1.076; no cell reaches
599 $T \geq 2$), and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells: all 51 cells with $T \geq$
600 1 are positive (mean +0.0195) and all 9 cells with $T < 1$ are negative (mean
601 -0.0047). This is the direction temperature scaling is expected to have — $T > 1$
602 softens an over-confident target and lowers its ECE, while $T < 1$ sharpens it and
603 raises ECE — and it supplies a unified mechanism for the nine counter-examples
604 of Sec. 6.1: they are exactly the nine cells with $T < 1$, including the extreme
605 CPSC→Chapman / ResNet1D / seed 44 case ($T = 0.932$, $\Delta\text{ECE}_{\text{OOD}} = -0.0168$),
606 the sharpest sharpening in the grid.

607 (iii) *Magnitude strata..* Under an additive floor the endpoint should be *smaller*
608 where the floor is larger. $\text{floor}_{\text{hi}}(T)$ reaches the mean effect only for $T \geq 1.25$ (4/60
609 cells); those cells give +0.0442 (4/4 positive) against +0.0138 (47/56 positive)

elsewhere. Restricting to the $T \geq 1$ branch — 51 cells, all positive, so the sign flip of (ii) cannot confound the comparison — and splitting at the median floor (0.0078) gives +0.0117 (26/26) below and +0.0276 (25/25) above: the endpoint is *larger* where the floor is larger, the opposite of an additive artefact. Since the floor attenuates, these values are lower bounds. This stratum comparison is nevertheless weaker evidence than (i)–(ii), because floor_{hi} is monotone in T on this side while the endpoint’s growth with T is partly mechanical.

(iv) *Negative control and additivity.* The independent ablation channel (same SmoothECE family) changes by -0.0124 on average and changes sign across cells (-0.083 to $+0.093$), so the metric has no systematic positive push. The floor at the cells’ own temperatures spans 0.0001 to 0.0595, up to $54\times$ the median per-cell bootstrap CI width (0.0011); an additive bias of that size would swamp the signal, yet subtracting the mid-band floor at each cell’s own temperature leaves $+0.0140$, i.e. 88% of the original $+0.0159$. The floor is therefore not a cell-level additive offset, and we do not subtract it per cell.

Consistency check. The endpoint also rises with T (OLS slope $+0.181$, $R^2 = 0.951$ over the 60 cells). We report this but do not rely on it: $T = 1$ is a fixed point at which the endpoint vanishes by construction, so part of this relation is mechanical. Its role here is that its sign is *opposite* to the floor’s contribution, which is the direction test of (i)–(ii).

Caveats. The floor is computed on idealised perfectly-calibrated relabellings of our own confidence distributions, so it is not a calibrated model of real ECE noise; the extreme stratum in (iii) has only 4 cells. The conclusion is exclusionary, not causal: the floor cannot account for the primary finding, but this does not by itself establish a causal mechanism.

635 *5.9. Discrimination-aware calibration gate (C5)*

636 Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for
637 deployment: a model whose discrimination has collapsed under shift may be re-
638 calibrable but clinically useless. We define a two-layer gate:

- 639 1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
- 640 2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-
641 class accuracy.

642 Across 12 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures), 7/12
643 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC,
644 CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (Incep-
645 tionTime only). The other 5/12 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-
646 XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls
647 below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The
648 strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but
649 $\text{acc} = 0.657 < 0.713$): TS improves calibration but discrimination has degraded
650 too far to be clinically actionable.

651 This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer
652 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes, Layer 2
653 rejects). It supplements C1 by also asking whether discrimination is sufficient for
654 the recalibrated predictions to be clinically useful.

655 *5.10. Real-data Shapley cross-fitting and predictability boundary*

656 The synthetic Shapley validation (C3) confirms decomposition recovery in the
657 identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx$
658 $\hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We cross-fit

all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

The linear model predicts the *direction* correctly in 51/60 (85%) but systematically *overestimates* the magnitude (mean prediction error +0.29, median +0.29). This is consistent with a predictability boundary: the decomposition captures the qualitative direction of the TS benefit but not its quantitative magnitude under real shift. The 9/60 direction errors cluster in CPSC→Chapman and Chapman→PTB-XL, where the Wasserstein-1 distance between source and target logit distributions is largest (mean 0.75 vs. 0.53 for correctly predicted directions), suggesting the linear approximation breaks down once the shift invalidates its underlying local Taylor expansion.

This strengthens the predictability failure branch (C3/C4): the Shapley decomposition is neither a valid *predictor* (*how much*) nor a valid *diagnostic* (*whether*). Its 51/60 direction accuracy coincides exactly with the constant-positive baseline (a model that always predicts “TS helps” scores 51/60 on this grid), so it carries no directional information beyond the prior that TS usually helps; the quantity that does recover the direction is the source-fitted temperature T (Sec. 5.5). For clinical translation, the gate in §5.9, using only the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the Shapley-predicted magnitude.

5.11. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60 (83.3%) of experiments, with mean gain +0.0027 and range $[-0.001, +0.017]$. The directions with the largest decision-cost reduction are CPSC→PTB-XL

684 (mean DCR +0.0625; net clinical value +0.0312) and CPSC→Chapman (mean
 685 DCR +0.0375; NCV +0.0188), which also have the largest E3-subsample
 686 $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.137 and +0.136 respectively, under the E3 subsampled
 687 binned-ECE protocol with $n_{\text{id}} = n_{\text{ood}} = 20$ patients). The direction with negative
 688 net benefit is Chapman→CPSC (mean DCR −0.0150; NCV −0.0075), which
 689 is correctly rejected by the discrimination gate (C5). We note that the decision-
 690 benefit ranking differs from the main-endpoint $\Delta\text{ECE}_{\text{OOD}}$ ranking (which peaks
 691 at PTB-XL→Chapman and PTB-XL→CPSC, means +0.0240 and +0.0238 on
 692 the full SmoothECE grid): the two protocols differ in estimator, class set, and
 693 subsample, so we treat the E3 analysis as a directionally supportive but protocol-
 694 distinct check, not a conversion of the primary endpoint. What survives is
 695 the gate-level conclusion: where the discrimination gate permits deployment,
 696 the calibration benefit does not cost decision performance; where it is negative
 697 (Chapman→CPSC), no calibration benefit should be deployed at all.

698 5.12. Net Clinical Value (NCV) results

699 The pre-registered NCV (Sec. 4.9) is BH-FDR corrected ($q = 0.05$) across the
 700 two secondary endpoints (DCR, NCV).

701 Result: the OOD NCV is +0.008125 with 95% CI [+0.0014, +0.0148]
 702 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS
 703 yields a net improvement in OOD clinical decisions. The ID NCV is −0.006292
 704 with 95% CI [−0.0114, −0.0012] ($p = 0.016$), *significantly negative*: TS slightly
 705 worsens ID decisions, consistent with the ID boundary condition (TS compresses
 706 confidence toward the base rate). OOD NCV Cohen’s $d = +0.31$ and ID NCV
 707 Cohen’s $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring

708 the main endpoint. Per-experiment NCV ranges over the 60 seeds are in the E3
709 CSV (`c1_dcr_ncv_multiclass.csv`).

710 5.13. Reliability Diagrams

711 Figure 5 shows the 6-direction averaged reliability curve, comparing model
712 confidence against empirical accuracy before and after temperature scaling. The
713 summary curve uses bin-count weighted averaging across all 6 transfer directions.

714 Figures 6–8 show the per-direction reliability diagrams for the representative
715 configuration (ResNet1D, seed 42). In each panel, the red squares show the raw
716 model calibration curve and the green circles show the post-TS calibration curve.
717 The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show
718 the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 re-
719 samples).

720 6. Discussion

721 6.1. OOD benefit seed variation and the nine counter-examples

722 The 51/60 robustness rate is driven by nine counter-examples across 5 of the
723 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL
724 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-
725 XL→CPSC has zero). We transparently disclose all nine rather than excluding
726 them, because the protocol requires all seed results to be reported and each has an
727 identifiable post-hoc mechanism supporting the qualified claim.

728 All nine share a single mechanism, stated in Sec. 5.8: they are exactly the nine
729 cells whose source-fitted temperature is $T < 1$. Because the temperature is fitted
730 on the source calibration split, $T < 1$ means the source model was *under*-confident

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

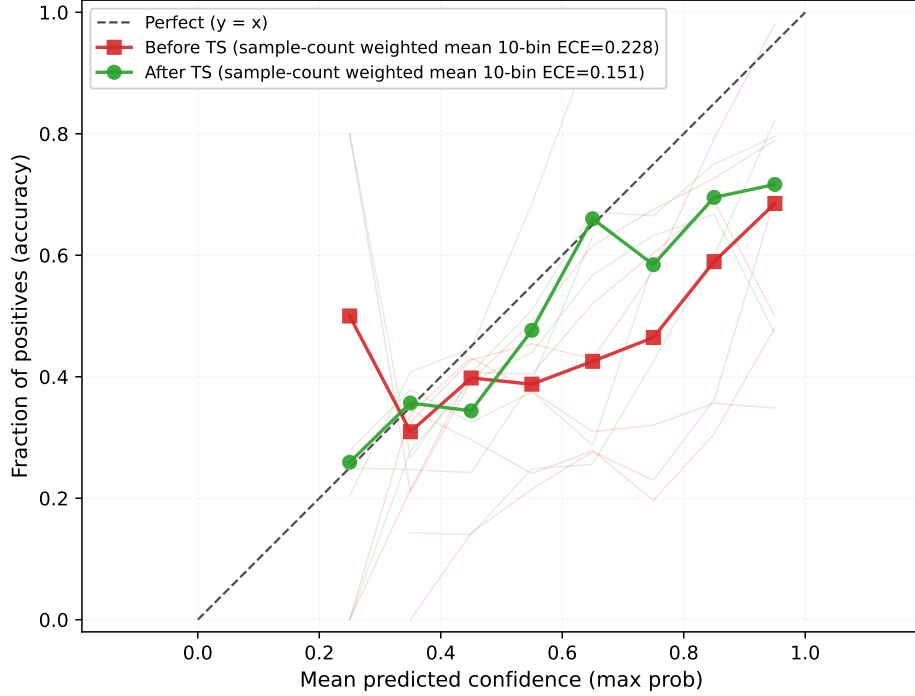


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

731 there; applying that temperature to the target sharpens an already over-confident
 732 distribution and raises its ECE. The counter-example set is therefore identified by
 733 a quantity available *before* the target is scored, and $\text{sign}(\Delta ECE_{\text{OOD}}) = \text{sign}(T - 1)$
 734 holds in all 60 cells. The per-cell accounts below give the accompanying discrim-
 735 inative and raw-ECE context; where a paragraph attributes the failure to “over-
 736 correction”, that is the $T < 1$ regime.

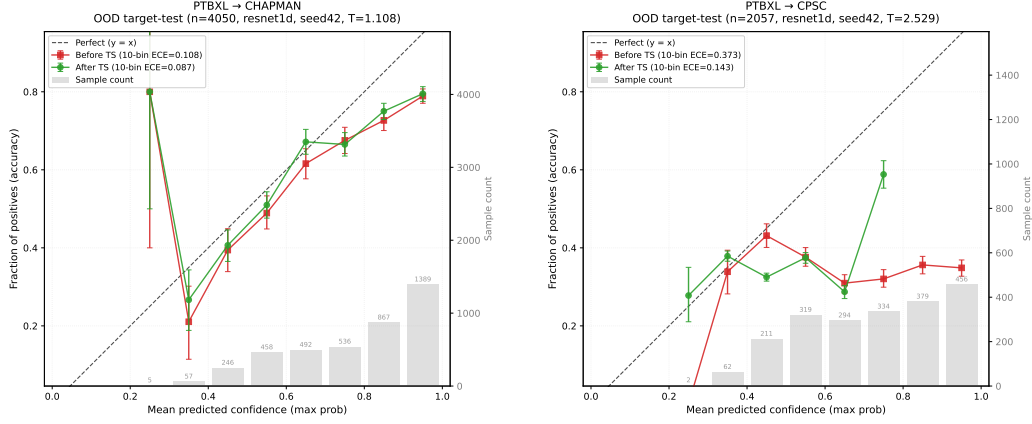


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

Counter-example 1: Chapman→CPSC / InceptionTime / seed 45.
 $\Delta ECE_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur:
 lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44,
 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature
 mismatch, not room shortage); and negative decay (-0.0075), the source-fit
 temperature amplifying rather than correcting residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} =$
 -0.0006 (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.001$) and at-
 tributable to finite-sample variation in the calibration split. The OOD raw ECE
 is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*); the mechanism is
 seed-level variation, not room shortage.

Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42.
 $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy is the low-
 est across the 5 seeds (0.4655) and $\Delta ECE_{\text{ID}} = +0.0004$ is near zero, so the
 temperature is well-tuned to ID but mismatched to the PTB-XL target.

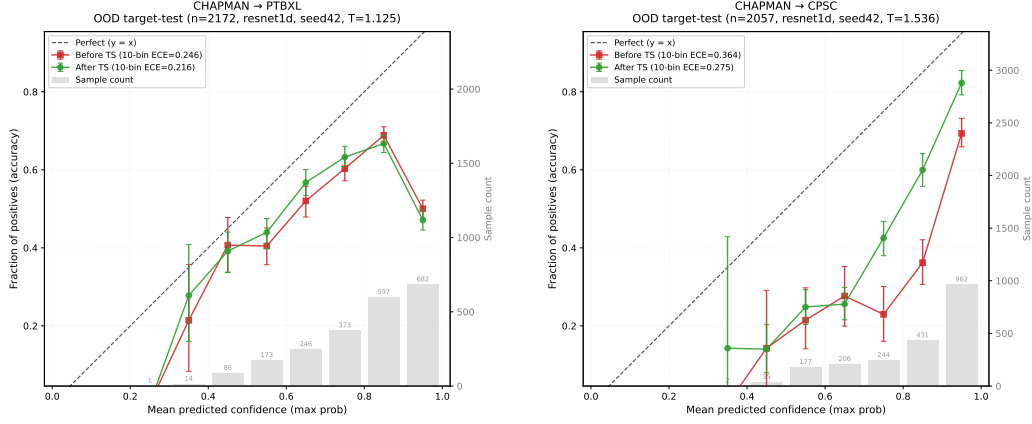


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and $\Delta ECE_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already well-calibrated on ID, leaving nothing to correct.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42. $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.002$); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE (0.0649), and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta ECE_{\text{ID}} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-

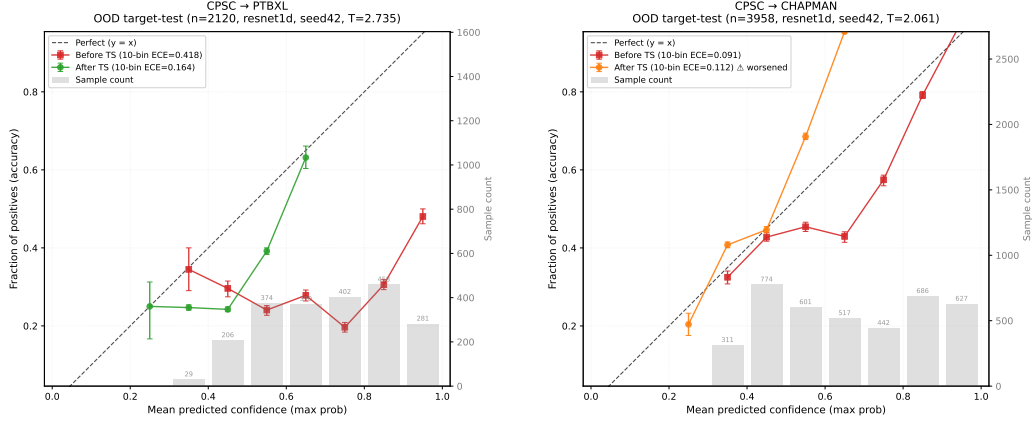


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

766 calibrated distribution degrades both ID and OOD calibration), unlike the “low
 767 OOD accuracy + large gap” mechanism of the other 8.

768 Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. $\Delta ECE_{OOD} =$
 769 -0.0050 (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest across the 5 seeds
 770 (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

771 Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. $\Delta ECE_{OOD} =$
 772 -0.0049 (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and
 773 $\Delta ECE_{ID} = -0.0032$ is slightly negative, leaving nothing to correct.

774 Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46.
 775 $\Delta ECE_{OOD} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983
 776 (lowest across the 5 seeds) and $\Delta ECE_{ID} = -0.0011$ is slightly negative, again
 777 leaving nothing to correct.

778 Common pattern: eight of nine counter-examples co-occur with (a) low OOD
 779 accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temper-
 780 ature mismatch; the ninth (CE6) is over-correction on an already well-calibrated

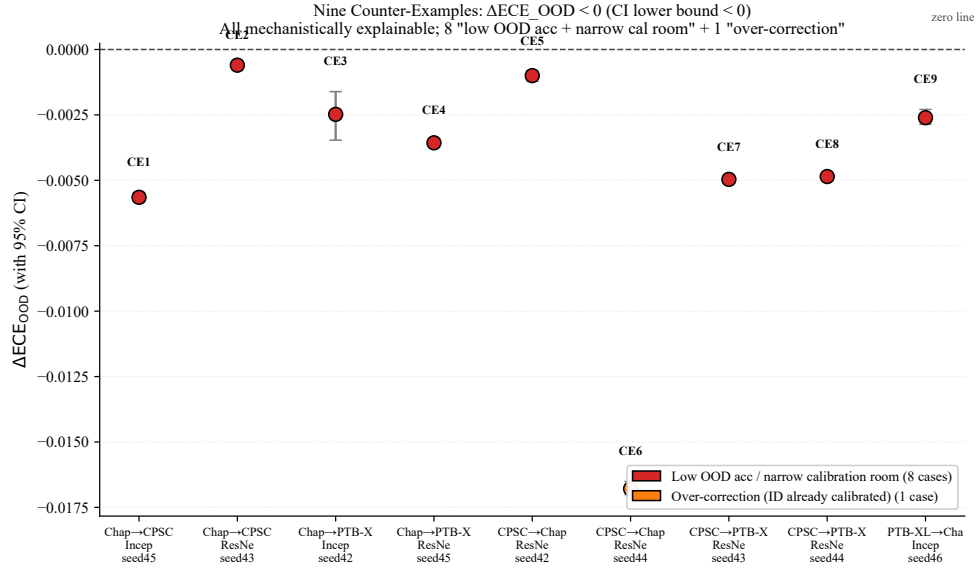


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

781 backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so
782 the failure mechanism is seed-level temperature mis-transfer, not calibration-room
783 shortage. The counter-examples are local, explainable random variation; the other
784 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

785 The 15.0% rate exceeds the 5% threshold common in clinical safety deploy-
786 ment. As a methodological boundary investigation, not a clinical deployment
787 guide, the relevant criterion is whether the effect is detectable and the mecha-
788 nism explainable, not a zero failure rate. All 9 counter-examples have identifiable
789 mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”), and the
790 expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

791 6.2. *Prior correction vs. posterior correction robustness*

792 The method-selection boundary (Table 3) reveals a robustness asymmetry
793 between prior-correction methods (EM, BBSE) and posterior-correction meth-
794 ods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30
795 and 24/30 support respectively with small variance; prior-correction methods
796 achieve higher mean ΔECE_{OOD} on the PTB-XL→CPSC pair (EM +0.0449,
797 BBSE +0.0777) but much higher variance on the Chapman→CPSC pair (EM CI
798 lower bound minimum −0.0763, BBSE −0.0720). The mechanism is the prior-
799 likelihood mismatch: prior-correction methods fit a target prior that, when the
800 source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibra-
801 tion. Posterior-correction methods do not estimate a target prior and are therefore
802 immune to this failure mode, at the cost of lower mean benefit when the prior
803 correction *would* work.

804 6.3. *Boundary condition: the ID benefit is small but nonzero*

805 The ID boundary condition (C2) has a direct interpretation under the triggered
806 branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing zero; the observed
807 rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a small but significant
808 positive ID benefit (mean $+0.0083$ vs. OOD $+0.0159$; stratum means -0.0008 to
809 $+0.0232$, Table 2). The boundary condition does *not* mean TS is ineffective on ID
810 nor that the ID benefit is zero: the ID benefit is on average about half the OOD
811 benefit, so part of the apparent OOD gain is shared with the ID domain. A val-
812 ECE early-stopping effect (model selection within the endpoint family) plausibly
813 contributes to the positive ID estimates, opposite to the protocol’s val-NLL rule (a
814 registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores
815 the decay contrast $= \Delta \text{ECE}_{\text{OOD}} - \Delta \text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is
816 qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively
817 OOD effect. The ID-vs-OD asymmetry is therefore a *difference in magnitude*
818 between two positive benefits, not presence versus absence.

819 6.4. *Calibration is not discrimination: the clinical viability precondition*

820 Temperature scaling is a monotone transform of the softmax probabilities;
821 it preserves the argmax by construction (asserted bitwise in the released sanity
822 check), so it cannot change classification accuracy, AUROC, or AUPRC. The pri-
823 mary endpoint $\Delta \text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes
824 an already-deployable model *trustworthy*; it cannot make a non-discriminative
825 model deployable.

826 The accuracy statistics make this concrete. Across the 60 TS experiments,
827 mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean

828 ID–OOD drop is 0.236 (maximum 0.412). Against each target corpus’s majority-
 829 class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), mean
 830 OOD accuracy is *below* baseline in 3 of 6 directions, with a fourth essentially
 831 at baseline: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411
 832 (0.013 above baseline); CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620
 833 (both well below 0.713). In those directions recalibration is moot, since no tem-
 834 perature improves a model that should be replaced by a majority-class predictor,
 835 consistent with the counter-example pattern (6/9 with low OOD accuracy). The re-
 836 leased artifacts include per-direction AUROC/AUPRC for all 60 experiments (re-
 837 sults/discrimination_metrics_60exp.csv): mean OOD AUROC ranges from 0.629
 838 (CPSC→Chapman) to 0.795 (PTB-XL→Chapman), i.e. far above the accuracy-
 839 based majority-class reading yet direction-dependent; and TS leaves AUROC un-
 840 changed (mean $\Delta\text{AUROC}_{\text{TS-raw}} = -0.001$, max |change| 0.024), as the monotone
 841 transform requires. The discrimination-first gate—measure target discrimination
 842 before investing in recalibration—precedes the calibration decision rule in deploy-
 843 ment.

844 6.5. *Explainable* $\neq$ *predictable*: post-hoc vs. pre-registered

845 We distinguish *explainable* from *predictable*:

- 846 • *Explainable* (post-hoc, mechanism-level): after observing a counter-
 847 example, we identify the co-occurring conditions (low OOD accuracy, low
 848 OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. 6.1),
 849 a post-hoc explanation, not a pre-registered prediction.
- 850 • *Predictable* (pre-registered, deployment-time): a target-domain unsuper-
 851 vised statistic (predicted-prior L_1 distance, logit moments) predicts the ben-

852 efit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 .
853 The predictability experiment (Sec. 5.5) triggers the failure branch (LOO
854 R^2 CI upper bound $0.104 < 0.5$).

855 The counter-examples are explainable, and they are also *locatable* in advance by
856 one target-label-free statistic: all nine are the cells with $T < 1$ (Sec. 6.1). What
857 fails is the stronger pre-registered requirement — a label-free feature that recovers
858 the *magnitude* of benefit retention with LOO R^2 CI upper bound ≥ 0.5 — which is
859 why the predictability experiment triggers the failure branch despite mechanistic
860 explainability. Note that T ’s sign rule is not a deployable criterion for deciding
861 whether to apply TS: it states the direction the transform must have by construc-
862 tion, not evidence that the transform helps (Sec. 5.8). The two-step chain (decom-
863 position + deployment lookup table) provides post-hoc explanation and actionable
864 guidance without a pre-registered predictability claim.

865 6.6. *Transparent re-location vs. HARKing*

866 The pre-registration revision A1 (primary endpoint re-location from decay to
867 $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [28, 27] is
868 whether the revision direction is theory-driven or result-driven. Our revision direc-
869 tion (“OOD is the calibration benefit source”) is consistent with the empirically
870 observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we
871 explicitly frame as an empirical prior rather than a theorem; the exploratory pilot
872 result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The
873 pilot results do not enter the main test report (protocol §7). All hypotheses in this
874 paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect
875 the post-revision status.

876 6.7. *Net clinical value: implications of the directional NCV asymmetry*

877 The NCV (Sec. 5.12) shows a directional asymmetry: OOD NCV =
878 $+0.008125$ ($p = 0.019$) is a small but significant positive value (recalibration im-
879 proves decisions on unseen target data more often than it degrades them); ID
880 NCV = -0.006292 ($p = 0.016$) is a small but significant negative value (a minor
881 decision-cost penalty on data the source model already calibrates well).

882 Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60
883 positive) is supported: the positive OOD NCV shows the calibration gain trans-
884 lates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost**
885 **improvement claim** (C5: 7/12 strata deployable) is partially qualified: the nega-
886 tive ID NCV tempers source-domain deployability. We therefore frame the NCV
887 as *supportive of OOD deployment* and *cautionary for ID deployment*. It also in-
888 forms the meta-analytic pooling (Sec. 7), where the asymmetry contributes to
889 CI-width heterogeneity across splits. A permutation test with tighter CIs and a
890 multi-threshold NCV analysis are registered as future work.

891 6.8. *Limitations and future work*

892 Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on
893 all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); Bi-
894 Mamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on
895 an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-
896 step to two-step); the deployment criterion is an empirical lookup table, not a
897 predictive criterion; (iii) real-data Shapley cross-fitting (§5.10) shows 85% di-
898 rection accuracy but systematic magnitude overestimation (mean error $+0.29$),
899 and a sensitivity analysis with T , Wasserstein-1, and entropy features does *not*
900 confirm a robust predictability boundary: only T predicts ($\text{LOO } R^2 = +0.34$,

901 and $\text{sign}(\Delta \text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells), so the retained claim is the
 902 weaker one that no label-free feature reaches $R^2 = 0.5$ under the implemented
 903 fold unit (Sec. 5.5); (iv) the 27-cell decomposition validation uses binned ECE,
 904 not the main endpoint’s SmoothECE, so quantitative conclusions cannot be ex-
 905 trapolated across ECE definitions; (v) the safety criterion is rarely satisfied on
 906 the full L2 shift matrix (TS safety rate 0.0064); TS is a useful default, not uni-
 907 versally safe; (vi) nine counter-examples of 60 experiments (15.0%) are trans-
 908 parently disclosed, with a frequent post-hoc pattern (low OOD accuracy, non-
 909 positive decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the op-
 910 erating CI is BCa at $B = 10,000$, re-run on all 60 experiments (60/60 success-
 911 ful, metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 ex-
 912 periments are retained as historical provenance and are no longer used for any
 913 reported number. All 60 main-grid cells carry complete $B = 10,000$ /BCa prove-
 914 nance; the only cells lacking such metadata are the two excluded BiMamba toy
 915 runs (PTB-XL→Chapman, seeds 42–43), which do not enter the main endpoint.
 916 (viii) the Youden threshold is fitted in-sample (the protocol’s held-out dose lev-
 917 els were not used), making the operating characteristic optimistic; (ix) the L2
 918 deployment/safety grid has since been completed on all 390 planned (pair, seed,
 919 shift) cells (`results/l2_shift_full_390cells.csv`), and the L2 noise levels are syn-
 920 thetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavail-
 921 able under PhysioNet licensing); (x) temperature was fitted once per experiment,
 922 so CIs omit temperature-fitting uncertainty (the pre-registered two-layer bootstrap
 923 is implemented but not enabled); (xi) model selection used validation ECE rather
 924 than the pre-registered NLL early-stopping rule, plausibly contributing to the pos-
 925 itive ID boundary estimates; (xii) the per-pair decomposition into covariate, prior-

926 probability, and concept-shift components is not asserted, since each pair mixes all
 927 three, isolated at eval time; (xiii) the deployment lookup table requires labeled tar-
 928 get data, and the label-free proxies failed their pre-registered branch, so no label-
 929 free gate is claimed; (xiv) TS is argmax-invariant and cannot improve accuracy,
 930 AUROC, or AUPRC; the artifacts now include per-direction AUROC/AUPRC
 931 for all 60 experiments (results/discrimination_metrics_60exp.csv), so the viabil-
 932 ity precondition is screened by both accuracy and discrimination (OOD AUROC
 933 means range 0.629–0.792 by direction), with OOD accuracy below the majority-
 934 class baseline in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is direc-
 935 tionally asymmetric (OOD = +0.008125, $p = 0.019$, positive; ID = −0.006292,
 936 $p = 0.016$, negative), qualifying the C5 claim for source-domain but supporting
 937 target-domain deployment; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$
 938 to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits,
 939 with 48 as future work; (xvii) this work is from a single-author study, limiting inde-
 940 pendent review and audit, partially mitigated by the pre-registration protocol and
 941 public repository; (xviii) the secondary endpoint G (recoverable loss versus an or-
 942 acle recalibrator) is *not delivered*: `fit_oracle/apply_oracle` are implemented
 943 but were never executed on real data, so G is undefined in this study and no G
 944 result is reported anywhere in this paper or the repository; (xix) no signal-quality-
 945 index (SQI) stratification is performed: the target corpora are pipeline-processed
 946 releases and the L2 noise levels are synthetic, so the calibration benefit is not
 947 characterised as a function of measured signal quality (pSQI/kSQI-type indices);
 948 this is the principal gap for a signal-processing readership and is the first item of
 949 future work; (xx) no test–retest or repeatability analysis of the fitted temperature
 950 is possible within this data regime: all three public corpora are cross-sectional

951 and provide a single recording per patient (Chapman-Shaoxing reports exactly
 952 one ECG per patient, §3), so same-patient across-session stability of T cannot be
 953 estimated from these data and is deferred to prospective multi-session collection.
 954 Future work: (1) prospective multi-device validation; (2) continuous shift spaces;
 955 (3) π -independent recovery estimators (BBSE/EM-class): BBSE is now delivered
 956 on the real grid (Honesty item 1), extension to further estimator classes remains;
 957 (4) the ResNet1D deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to
 958 test for a systematic interaction; (5) confirmatory testing under the A2 reduction
 959 (12 hypotheses, BH-12 $q = 0.05$, 9/12 significant, Bonferroni-12 = 2/12); (6) re-
 960 running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU;
 961 (7) a multi-threshold permutation test on NCV.

962 7. Conclusion

963 We pre-registered and validated a multi-seed calibration boundary study of
 964 temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD benefit: 51/60
 965 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime 27/30 = 90.0%,
 966 ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID boundary: 23/60 cells
 967 (38.3%) have 95% CIs containing zero under the BCa operating CI (percentile:
 968 29/60 = 48.3%), below the $\geq 80\%$ criterion, triggering the pre-registered branch;
 969 the OOD benefit is $1.9\times$ the ID benefit. **(C3)** Shapley decomposition: 27/27 cells
 970 pass at $n = 20,000$ (21/27 at $n = 500$). **(C4)** Method selection: TS robust (51/60),
 971 EM prior and Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/12 strata pass both
 972 layers, reframing the 9 counter-examples as 4 calibration and 5 discrimination
 973 failures. The predictability failure branch is triggered (three-step chain becomes
 974 two-step), and all hypotheses are exploratory + robustness validation, not confir-

975 matory; protocol v2.1-A1 applies. TS needs local validation (safety rate 0.0064),
 976 and OOD accuracy is below the majority-class baseline in 3 of 6 directions.

977 A noise floor control (§5.7) bounds, but does not eliminate, a SmoothECE ar-
 978 tifact: at moderate temperatures metric noise is $2\text{--}21\times$ below the observed raw
 979 ECE (0.036–0.357, mean 0.211), whereas for large $T > 3$ the floor (0.10–0.16) is
 980 comparable to the observed $|\Delta\text{ECE}|$ and a metric-noise contribution cannot be ex-
 981 cluded there. Real-data Shapley cross-fitting (§5.10) gives 85% direction accuracy
 982 with magnitude overestimation (mean error +0.29). The predictability boundary
 983 holds only in the weak sense that no label-free feature reaches $R^2 = 0.5$ under the
 984 implemented fold unit: the source-fitted temperature T does predict (+0.34, and
 985 the sign of $\Delta\text{ECE}_{\text{OOD}}$ in 60/60 cells), while Wasserstein-1 and entropy features do
 986 not.

987 The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative
 988 (-0.000573) because narrow-CI studies dominate, while DerSimonian–Laird ran-
 989 dom effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossAr-
 990 chitecture +0.0148). We caution, however, that the meta-analytic independence
 991 assumption is *not* satisfied here: the 60 cells share the same three corpora,
 992 patient sets, and architectures, and their between-cell Q statistics are extreme
 993 ($I^2 \approx 99.98\%$), so both FE and DL confidence intervals are spuriously narrow.
 994 The interpretable pooled uncertainty is therefore the cluster-robust interval over
 995 the $12 \text{ pair} \times \text{architecture}$ strata: $\Delta\text{ECE}_{\text{OOD}} = +0.0159$ [0.0113, 0.0205] (t, df= 11;
 996 stratum SD 0.0072, about $1800\times$ the single-experiment bootstrap SE). The point
 997 estimate 0.0148–0.0159 and the count-based proportion ($51/60 = 85.0\%$) stand.

998 The 60 experiments form the robustness-evidence family; amendment A2 re-
 999 duces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypothe-

ses ($6 \text{ pairs} \times 2 \text{ architectures} \times 1 \text{ primary shift level}$), with all 13 L2 shift levels re-designated as exploratory dose-response. BH-60 and Bonferroni leave the robustness headline unchanged (51/60); the confirmatory family is BH-12 with 9/12 significant, and Bonferroni-12 retains 2/12. Across the 60 TS experiments the 95% CI widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall below the narrow-CI diagnostic threshold (0.005) and six supporting cells have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60 (75.0%). Narrow CIs reflect the moderate calibration sample and low SmoothECE variance; they do not affect the sign-based support/counter-example classification.

1009 **Honesty Declaration (5 items, adversarial audit product)**

1010 1. MAPE_π is a **design check** (resampling constant read-back), not an estimator
 1011 capability test. The independent π recovery registered as deferred (protocol
 1012 §13, item 7) has since been delivered on the real 60-cell grid: with target la-
 1013 bels withheld, BBSE reduces the prior-recovery error from 0.1916 (source-
 1014 prior baseline) to 0.1132 — a 40.9% relative reduction, better than the base-
 1015 line in 56/60 cells, positive in 6/6 transfer pairs, with a pair-clustered boot-
 1016 strap CI of $[+0.048, +0.111]$. EM gives no improvement (0.1923). Two
 1017 limits remain and are reported as such: the absolute error is still 0.113, and
 1018 the confusion-matrix condition number exceeds 10^3 in 5/60 cells and 10^4
 1019 in 1/60, so recovery is numerically boundary-sensitive there. **This does**
 1020 **not repair the decomposition estimator**, which targets a different quantity
 1021 (the target class prior, not the attribution coefficients) and whose real-data
 1022 cross-fitting error remains +0.29 (Sec. 5.10).

- 1023 2. Shapley shares in the `share_reliable=False` regime are **not inter-**
1024 **pretable as shares**; ranking uses absolute- value CIs.
- 1025 3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the
1026 archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information
1027 mode), reported via exit code 2.
- 1028 4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
1029 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement
1030 demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13).
1031 The wording is limited to “mechanism decomposition (synthetic validation
1032 part)”.
- 1033 5. The repository’s `scripts/preprocess_cinc2021.py`,
1034 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
1035 from the third-party SafeECGMatch release (original README in
1036 `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment
1037 grid.

1038 Data availability

1039 The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018 (CPSC Database
1040 and CPSC-Extra) [5, 6] datasets are publicly available via PhysioNet un-
1041 der their respective licenses. The pre-registered protocol v2.1-A1, its hash
1042 manifest (`docs/osf_archive_manifest.json`), the revision A1 docu-
1043 ment, and the full per-experiment result files (`checkpoints/transfer/**/`
1044 `transfer_result.json`, `results/*.csv`) are included with the submission
1045 as supplementary material and are also available in the public repository
1046 <https://github.com/Toni-cs/ecg-calibration-transportability>.

1047 The protocol, both amendments (A1, A2), and a per-file archival-status re-
1048 port are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
1049 docs/osf_archive_manifest.json is the internal snapshot taken on 2026-09-
1050 05; it is not a substitute for the public record.

1051 **Code availability**

1052 The full pipeline code, preprocessing (reproducible from the public source
1053 databases), patient-level split generation, model training, the calibration-method
1054 implementations, and all statistical analyses, together with the environment
1055 lock file, the patient-level split indices for all five dataset variants and five
1056 random seeds, and selected per-experiment result artifacts, is publicly available
1057 at <https://github.com/Toni-cs/ecg-calibration-transportability>.
1058 Trained model checkpoints for the 60-experiment transfer grid are provided as
1059 release assets of the same repository. The repository will be archived on Zenodo
1060 with a DOI upon acceptance. No new raw waveforms were collected or generated;
1061 the patient-level split indices and label-mapping tables released with the code
1062 fully determine the data configuration used in every experiment.

1063 **Declaration of generative AI and AI-assisted technologies in the writing pro-** 1064 **cess**

1065 All scientific work reported in this study—the study design, the pre-registered
1066 protocol and its amendments, the implementation and execution of the 60-
1067 experiment transfer grid, the statistical analysis, and the interpretation of the
1068 results—was performed by the author. A large language model assistant (GLM)
1069 was used in two supervised roles, neither of which produced scientific content:

1070 as a coding assistant that drafted and refactored parts of the analysis and plotting
1071 code (Section 4.10), and as a language-editing aid that improved the wording and
1072 readability of text the author had written. The tool generated no experimental data,
1073 results, or conclusions, and made no analytical or interpretative decisions; every
1074 reported number is produced by the author's scripts and is reproducible from the
1075 public repository. After using the tool, the author reviewed and edited all affected
1076 code and text as needed and takes full responsibility for the content of the pub-
1077 lished article.

1078 **CRedit authorship contribution statement**

1079 **Toni Guan:** Conceptualization, Methodology, Software, Validation, Formal
1080 analysis, Investigation, Data curation, Writing – original draft, Writing – review
1081 & editing, Visualization, Project administration.

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1089 **Declaration of competing interest**

1090 The author declares no competing financial or personal interests.

1091 The 60 per-seed reliability diagrams ($6 \text{ transfer pairs} \times 2 \text{ architectures} \times 5$
1092 seeds; calibration curves, reliability diagrams, and ΔECE bars per experiment)
1093 are provided as a separate supplementary file (supplementary.pdf, §1) and are
1094 additionally released with the code repository.

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